

Bi-LSTM-Based Trading Strategy for Long-Term Profit Optimization in Thailand's SET50 Index Using Multi-Source Data

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Introduction

The rapid development of machine learning has transformed financial market prediction, enabling the establishment of data-driven techniques that adapt to complex market dynamics. The SET50 of Thailand, consisting of the 50 most liquid and high-capitalized stocks, serves as an important benchmark for investors and fund managers seeking opportunities in the Thai stock market. With the development of algorithmic trading, there is a growing interest in complex models that can leverage data to support long-term investment strategies. This project aims to investigate the implementation of deep learning strategies, specifically Bi-Directional Long Short-Term Memory (Bi-LSTM), to forecast trends in the SET50 Index and improve trading strategies over extended periods.

Machine learning algorithms have gained significant attention in stock market forecasting due to their ability to identify complex, non-linear relationships in financial data. Traditional statistical methods struggle to capture temporal relationships and adapt to volatile stock market patterns. In contrast, deep learning models such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) have demonstrated superior performance in capturing sequential data such as stock prices. Bi-LSTM, an extension of LSTM, improves this ability by analyzing input data in a forward and backward directions, offering a more comprehensive understanding of market trends and the potential to significantly improve prediction accuracy.

Despite the accessibility of a variety of financial data and the capabilities of Bi-LSTM, their combined application in the context of Thailand's SET50 Index is still understudied, especially for long-term profit optimization strategies. Most of the current research focuses on short-term price changes or uses limited data types, which can weaken the robustness in real-world situations. This study addresses that gap by integrating Bi-LSTM with multi-source data to develop a predictive model tailored to the SET50 Index. By evaluating the model's performance using both financial return metrics and prediction accuracy, this study aims to provide an efficient and flexible framework for long-term profit optimization development strategies in emerging markets.

Problem statement/Research Question(s)

Although machine learning is now recognized as an effective tool in financial forecasting, most of the current models have been developed for short-term trading and rely on single-source data. Research on long-term investment strategies using advanced deep learning methods and multi-source data integration within the framework of Thailand's SET50 Index is limited. Research usually uses traditional statistical methods or focuses on daily price shift forecasting, which may insufficiently capture comprehensive market trends essential for sustained profitability.

This project addresses an important gap by exploring the application of Bi-Directional Long Short-Term Memory (Bi-LSTM) models to develop robust trading strategies focused on long-term profit maximization in the SET50 Index. This study aims to develop and evaluate a forecasting model that is more adaptive and predictive over long-term investment periods by combining several data sources, including technical indicators, price history, and sentiment signals. The findings will benefit individual investors, financial analysts, and algorithmic trading systems in enhancing decision-making in emerging markets especially Thailand.

Research Question:

- How can a Bi-LSTM model trained on multi-source financial data be used to optimize long-term trading profitability in Thailand's SET50 Index?

Literature review

Machine Learning in Financial Trading

Machine Learning techniques have become a key instrument in modern financial trading. To recognize patterns in stock market data and forecast future trends, traditional methods like Support Vector Machines (SVMs), Random Forests, and Artificial Neural Networks (ANNs) have been used. (Atsalakis & Valavanis, 2009; Ajiga et al., 2024). These models are effective for short-term predictions and can process large datasets with many features, often outperforming statistical methods in terms of accuracy (Preeti et al., 2018).

However, these traditional ML models typically fail to process long-term temporal dependencies, which are necessary for investment strategies in long periods of time. Due to their limited memory and sequential reasoning capabilities, models like SVMs or Random Forests have trouble capturing dynamic relationships that change over time (Fischer & Krauss, 2018). Because deep learning algorithms like Long Short-Term Memory (LSTM) and Bidirectional LSTM (Bi-LSTM) can maintain pattern recognition throughout

longer sequences of data, they are better suited for time-series data. This has led us to study these algorithms.

The lack of these complex deep learning algorithms for long-term financial forecasting in emerging markets is our research gap. Few studies have explicitly applied Bi-LSTM to comprehensive trading strategies that integrate multi-source data, like stock prices, technical indicators, and sentiment signals, for long term profit optimization, even though it has demonstrated strong performance in a variety of domains (Chaweewanchon and Chaysiri, 2022). This research aims to close that gap.

Bi-LSTM for Time-Series Forecasting

Bidirectional Long Short-Term Memory (Bi-LSTM) is a type of LSTM algorithm that is designed to learn from both past and future contexts within time-series data. This algorithm enhances the model's ability to capture complex dependencies, which is advantageous in financial time-series forecasting where market dynamics are influenced by both short-term variation and longer-term historical trends. Unlike others traditional ML methods, Bi-LSTM has been demonstrated to improve prediction accuracy by capturing patterns in both short and long term (Siarni-Namini et. al., 2019).

In past studies, Bi-LSTM models have outperformed other deep learning models such as standard LSTM, RNN, and CNN in forecasting financial information. For example, Chaweewanchon and Chaysiri (2022) demonstrated that combining Bi-LSTM with CNN and robust feature selection significantly enhance stock price prediction accuracy in the SET50 Index. Similarly, Preeti et al. (2018) proved that deep leaning networks, when compared with classical statistics methods like Auto Regressive Integrated Moving Average (ARIMA) or General Regression Neural Networks (GRNN), often have considerable improvements in prediction precision because of their ability to model nonlinear, non-stationary sequences. These findings suggest that Bi-LSTM is well-suited for handling the complexity and volatility of financial markets, especially when used with multi-source datasets. As a result, Bi-LSTM is progressively recognized as a powerful tool for developing data-driven trading strategies with long-term forecasting.

Multi-Source Data in Trading Strategies

Integrating multi-source data in trading strategies has become a crucial approach for enhancing the prediction accuracy of financial models. Traditional statistics methods often mainly rely on historical price data sources, but recent algorithms demonstrate that combining multi-source data such as technical indicators, price history, and sentiment signals can significantly improve model accuracy and robustness (Ajiga et al., 2024). This change demonstrates the complexity of modern financial markets, where investor behavior and asset

prices are influenced not just by past trends but also by news, economic reports, and social activities.

Several studies highlight the benefits of multi-source data integration in machine learning models. For example, Nelson et al. (2019) combined technical indicators like Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD) into LSTM-based algorithm, resulting in improved forecasting accuracy over just price input alone. Similarly, Chaweewanchon and Chaysiri (2022) used feature selection techniques to identify the most relevant input features, enhancing portfolio performance through predictive modeling of the SET50 Index. The effectiveness of these models is their ability to extract complex signals from various data domains such as technical, behavioral, and fundamental which when fed into deep learning models such as Bi-LSTM, provide a more extensive view of market dynamics. These findings support the idea that using multi-source data is not just additive, but it is essential to develop robust and adaptive trading strategies.

The SET50 Index

The SET50 Index, maintained by the Stock Exchange of Thailand (SET) including of the 50 most liquid and high-market-capitalization stocks in the Thai stock market. It serves as a crucial benchmark for institutional investors and algorithmic trading systems operating in Thai market. This index is dynamic and it's representing the behavioral of Thai market. For researchers, this index offers both data availability and market relevance, making it ideal for training and evaluating models over realistic conditions.

Compared to major indices in developed markets such as the S&P 500 or FTSE 100, the SET50 Index provides a relatively understudied opportunity in academic literature. While several studies have used models in Western or Chinese markets, few have applied advanced deep learning model, especially Bi-LSTM, on Thai market using multi-source data (Chaweewanchon and Chaysiri, 2022). This gap encourages research into how forecasting methods may be adapted for Southeast Asian financial contexts, where behavioral and structural dynamics may vary significantly.

Objective(s)

- To design and build a trading strategy using a Bi-LSTM neural network that combines multi-source data (e.g., price history, technical indicators, sentimental data) to predict future trends in Thailand's SET50 Index
- To test the accuracy and performance of the model by comparing its predictions and trading results with standard models or baseline strategies over a ten-year period.

Methodology

• Materials and Equipment

- SET50 historical data from Yahoo Finance
 - OHLCV (Open, High, Low, Close, Volume)
- Technical Indicators
 - Simple Moving Average (SMA)
 - Exponential Moving Average (EMA)
 - Relative Strength Index (RSI)
 - Moving Average Convergence Divergence (MACD)
 - Bollinger Bands
 - Stochastic Oscillator
- Sentiment Data
 - Thai financial news headline
 - Google News API
- Macroeconomic indicators
- Local computer with Python 3.11 on Visual Studio Code
- Google Collab
- Library and Frameworks
- Evaluation Techniques
 - Prediction Accuracy
 - Root Mean Square Error (RMSE)
 - Mean Absolute Error (MAE)
 - R-square (R^2)

• Procedure

1. Data Collection and Preprocessing
 - Collect all 50 stocks OHLVC data over a 10-year span
 - Program technical indicator (e.g., SMA, EMA)
 - Extract sentiment data using keyword-based scoring
 - Include macro features (e.g., exchanged rates, global indices) to the models
2. Feature Engineering and Sequence Construction
 - Align and normalize feature by using z-score and rolling normalization
 - Construct time series sequences of fixed window size (e.g., 30, 60, or 90 days)
 - The model will learn to predict future virtual trend vectors
3. Model Architecture and Learning Objective
 - Implement Bi-LSTM using TensorFlow/Keras

- The output will be continuous-valued multi-step return forecasts, not just directional classes.
- Include loss function such as Huber loss, Smooth L1, or custom loss to balancing short-term and long-term error.
- 4. Virtual Trading Strategy Simulation
- 5. Backtesting and Performance Evaluation
 - The strategy will be backtested over multiple timeframes (e.g., 2014–2021 for training, 2022–2024 for testing).
 - Evaluation metrics will include:
 - Cumulative Return
 - Sharpe Ratio and Sortino Ratio
 - Maximum Drawdown (MDD)
 - Profit Factor
 - Win Rate vs. Lose Rate.
 - Trade Frequency and Average Holding Period
- 6. Hyperparameter and Strategy Optimization
 - Both model parameter and strategy parameters will be optimized using
 - Grid search or Bayesian optimization
 - Walk-forward validation on rolling time window
- 7. Comparative Study and Baseline Models
 - The Bi-LSTM strategy will be benchmarked against:
 - Traditional machine learning models (e.g., Random Forest, XGBoost)
 - Technical-rule-based strategies (e.g., MA crossover, RSI signals)
 - Passive buy-and-hold strategy on the SET50 Index
 - Statistical significance tests (e.g., paired t-test on returns) will be conducted to assess superiority.
- 8. Real-World Applicability and Risk Management
 - The final output will be shown how the proposed model can be used for semi-automated trading and how it can integrate with real trading platforms (e.g., streaming SET data with APIs)
 - A risk analysis will be included, identifying weaknesses such as overfitting, latency, and sudden market shocks.

Expected outcome(s)

This project is expected to effectively improves long-term profit optimization in Thailand's SET50 Index using a Bi-LSTM model trained on multi-source financial data. By addressing the identified gap, which are the lack of deep learning applications for long-term trading in emerging markets, the proposed system aims to provide an advanced forecasting framework that extends beyond short-term prediction and toward actionable investment strategies over multi-week or multi-month range.

The project will demonstrate that combining price history, technical indicators, and sentiment data enhances the performance of deep learning models in predicting volatile and

complex markets like Thailand's SET50 Index. The prediction accuracy and the profitability, drawdown control, and risk-adjusted returns of the model will be validated by the virtual trading simulation and back testing process. This positions the model as a practical foundation for developing trading tools or decision support systems in local financial markets.

The key beneficiaries of this research will include:

- **Retail investors in Thailand**, who can gain access to more intelligent long-term investment strategies.
- **Financial analysts and quant researchers**, who may adopt the methodology or refine it with new data sources.
- **Educational institutions**, which can use this project as a case study in machine learning applications in finance.
- **The broader academic and algorithmic trading community**, by contributing a novel, Thailand-focused Bi-LSTM strategy to the literature.

Ultimately, this study illustrates the practical application of advanced machine learning in emerging market environments and lays the foundation for the future development of automated and personalized trading systems in Southeast Asia.

Work plan timeline

Methods	Month											
	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr
Choose Project Topic												
Literature Review												
Find Data												
Train & Test Model												
Improve Model												
Conclusion												
Writing Report												

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☐ Approved

☐ Revision required

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